

MUT ICT DEPARTMENT WEBSITE

Smart & Interactive Web Application

Module:

Internet Programming 2 (ITPR300/ITNP300)

Assessment:

Assessment 2

TEAM MEMBERS

Student Number	Surname	Initials	Role	Contribution
22440270	Mbhense	AN	Lead Developer	40%
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ABSTRACT

The MUT ICT Department Website is a smart and interactive web application designed to modernize the current ICT department website at Mangosuthu University of Technology. The existing website had several limitations including static content, lack of interactivity, no database integration, poor mobile responsiveness, and absence of intelligent features.

The proposed solution introduces a full-stack web application developed using HTML, CSS, JavaScript, Node.js, Express.js, and MySQL. The system includes user registration and login, AI-powered chatbot assistance, smart search functionality, campus tour booking, contact forms, responsive design, and CRUD operations.

The project also implements middleware for logging, validation, and error handling. UXD principles were applied to improve usability, accessibility, consistency, and responsiveness across devices.

The final system successfully demonstrates frontend interactivity, backend development, database integration, AI implementation, and modern web development practices.

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INTRODUCTION

Background

The ICT Department website at Mangosuthu University of Technology provides information about programmes, admissions, and departmental activities. However, the current website lacks modern functionality and interactivity expected from contemporary web systems.

Students cannot register or log into the platform, information is static, and there are no intelligent systems to assist users. Additionally, the website is not fully optimized for mobile devices.

Aim of the Project

The aim of this project is to develop a smart and interactive web application that improves user experience, introduces intelligent features, and modernizes the MUT ICT Department website.

Objectives

The objectives of the project are:

- To redesign the website using UXD principles**
- To create a responsive frontend interface**
- To develop a Node.js and Express.js backend**
- To integrate a MySQL database**
- To implement CRUD operations**
- To add AI-powered chatbot functionality**
- To improve website interactivity and accessibility**

PROBLEM STATEMENT

Current Website Weaknesses

Weakness	Description
Static Content	The website only displays fixed information with no dynamic updates
No User Accounts	Students cannot register, log in, or have personalized experiences
No Database Storage	Information is not stored persistently
No Interaction	Users cannot contact the department or book services online
No AI Assistance	No chatbot or intelligent features
Poor Mobile Experience	Website is not fully responsive
No Feedback System	Users receive no confirmation after form submissions

Proposed Solutions

Area	Improvement Implemented
User Accounts	Registration and login system with password hashing
Data Storage	MySQL database integration
Interactivity	Dynamic content, modal windows, validation
Communication	Contact forms and campus tour booking
AI Features	AI chatbot and smart search
Mobile First	Responsive design and hamburger menu
User Feedback	Success and error notifications

UXD ANALYSIS & FRONTEND REDESIGN

UX Weaknesses Identified

Weakness	Impact
Inconsistent navigation	Users become confused
No mobile responsiveness	Poor mobile usability
Poor color contrast	Accessibility problems
No form feedback	Users unsure if actions succeeded
Cluttered layout	Difficult navigation

UXD Principles Applied

Principle	Implementation
Consistency	Same layout and navigation across pages
Feedback	Success and error messages
Accessibility	High contrast colors and keyboard support
Readability	Clean typography
Responsiveness	CSS Grid and Flexbox

Navigation Structure

Home

- About
- Programmes
- Admissions
- Contact
- Search

Color Scheme

Color	HEX Code	Usage
Navy	#7B1C2E	Header, buttons
Gold	#C9A84C	Highlights
White	#FFFFFF	Background
Off-white	#FDF6F0	Alternate backgrounds

Typography

Font	Usage
Outfit	Headings
Crimson Pro	Body text

Responsive Breakpoints

Device	Breakpoint	Layout
Mobile	<768px	Single column
Tablet	768px–1024px	Two columns
Desktop	>1024px	Full layout

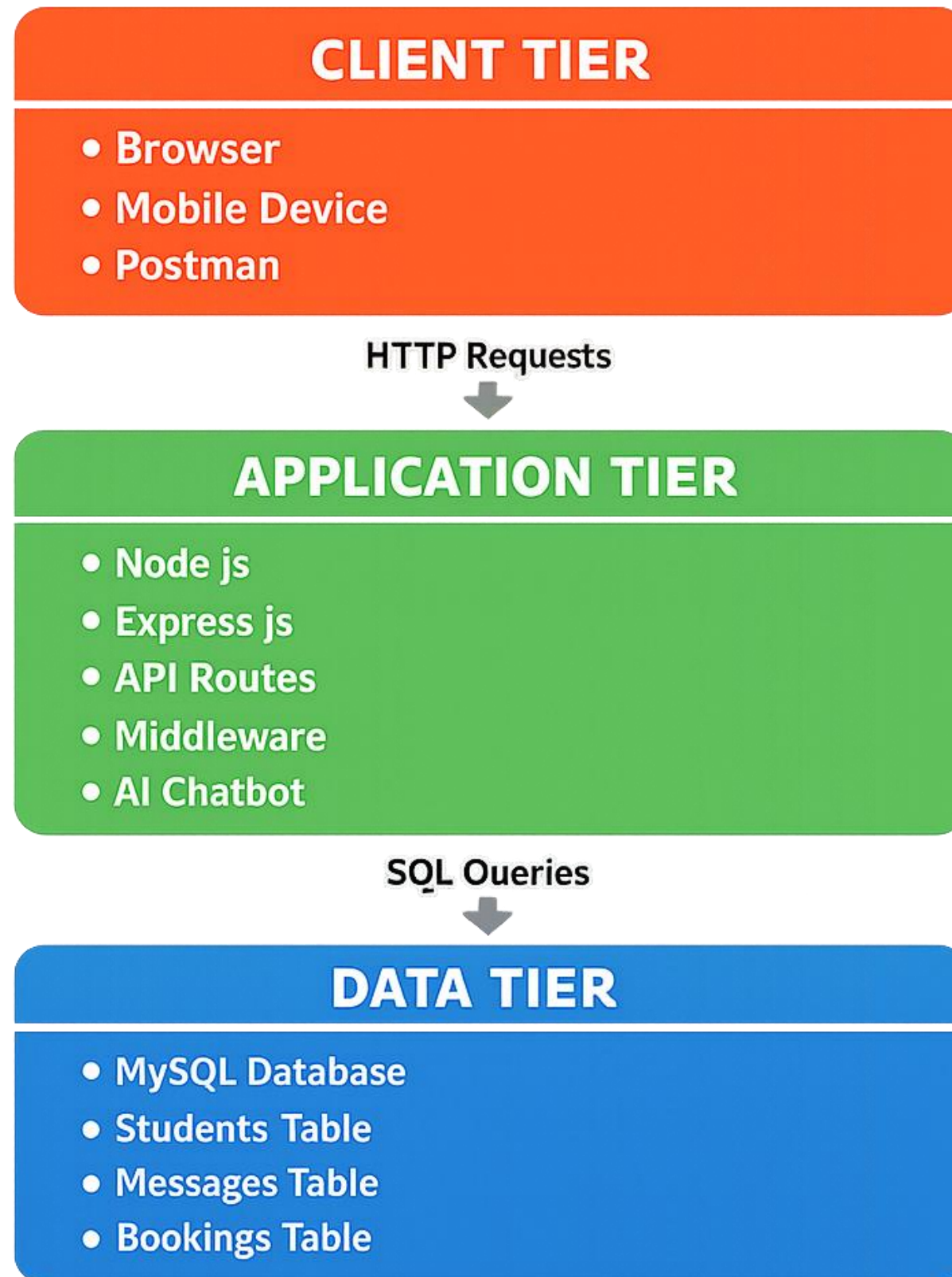
Dark Mode Implementation

The system includes a dark mode toggle that:

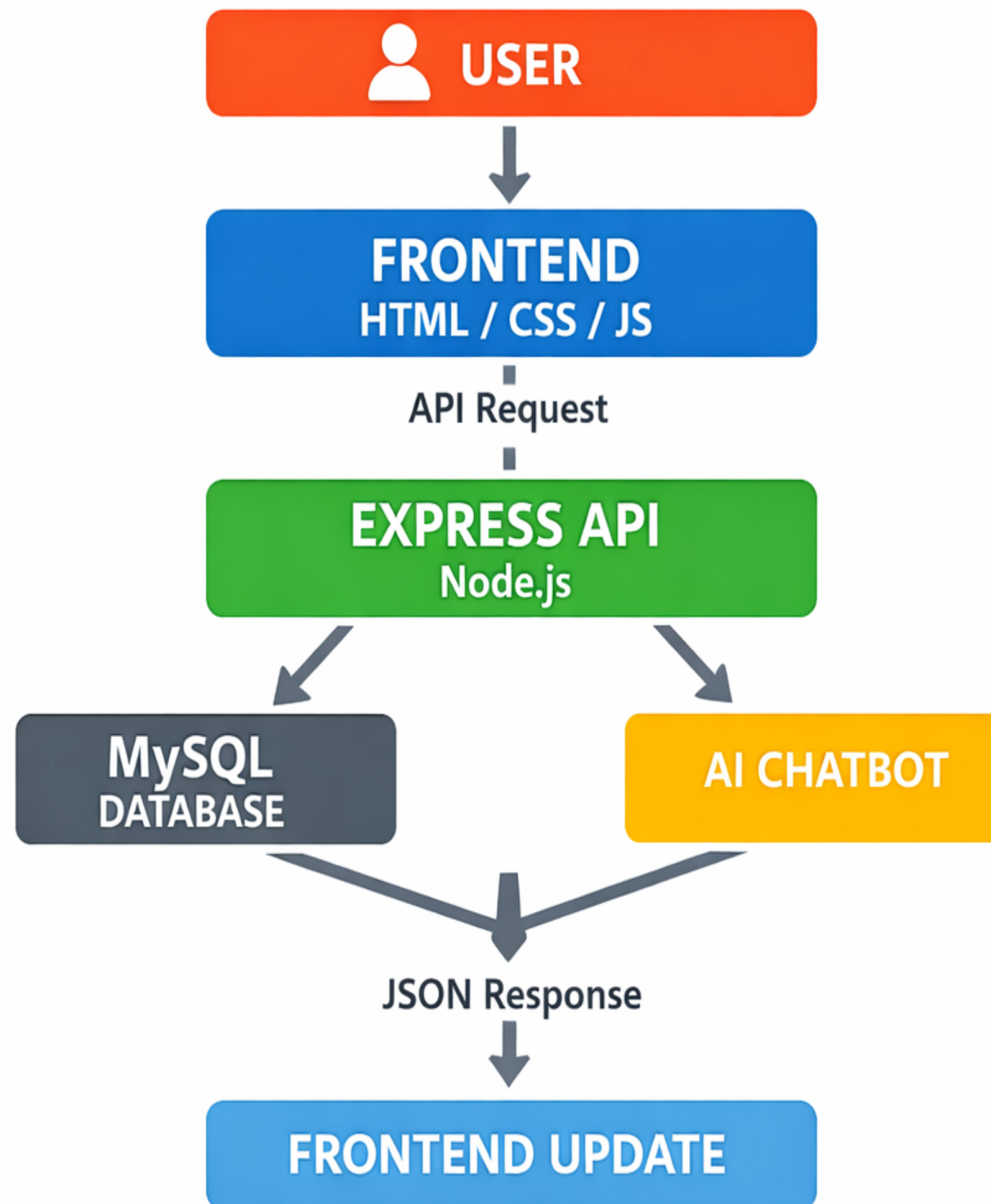
- Switches themes
- Saves preferences using localStorage
- Improves accessibility

SYSTEM ARCHITECTURE

Three-Tier Architecture



Data Flow



DATABASE DESIGN AND INTEGRATION

Database Name

Column	Type	Description
id	INT	Primary key
student_number	VARCHAR(20)	Student number
first_name	VARCHAR(100)	First name
last_name	VARCHAR(100)	Last name
email	VARCHAR(100)	Email
phone	VARCHAR(15)	Phone
course	VARCHAR(100)	Programme
year_level	INT	Year level
password_hash	VARCHAR(255)	Hashed password

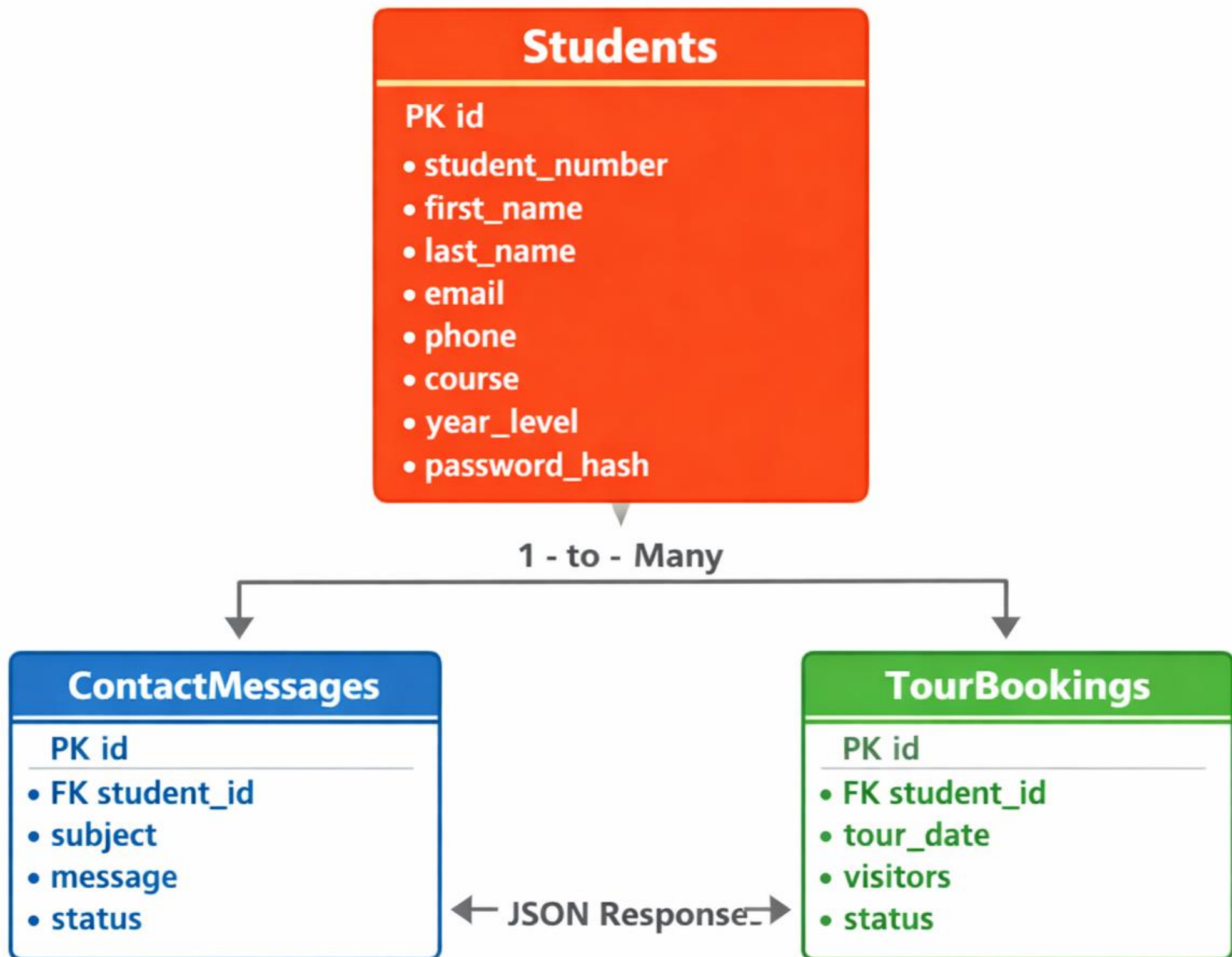
ContactMessages Table

Column	Type	Description
id	INT	Primary key
student_id	INT	Foreign key
subject	VARCHAR(200)	Message subject
message	TEXT	Message content
status	ENUM	Message status

TourBookings Table

Column	Type	Description
id	INT	Primary key
student_id	INT	Foreign key
tour_date	DATETIME	Tour date
visitors	VARCHAR(50)	Visitor count
status	ENUM	Booking status

Entity Relationship Diagram



BACKEND DEVELOPMENT

Technology Stack

Component	Technology
Runtime	Node.js
Framework	Express.js
Database Driver	mysql2
Password Security	bcrypt
AI Integration	OpenRouter API

API Endpoints

Method	Endpoint	Description
GET	/api/hello	Test server
POST	/api/register	Register student
POST	/api/login	Login
POST	/api/contact	Contact form
POST	/api/chat	AI chatbot
PUT	/api/student/:id	Update profile
DELETE	/api/booking/:id	Delete booking

Registration Route

JavaScript code:

```
app.post('/api/register', async (req, res) => {  
    // Registration logic  
}]);
```

Login Route

JavaScript code:

```
app.post('/api/login', async (req, res) => {  
    // Login logic  
}]);
```

MIDDLEWARE IMPLEMENTATION

Logging Middleware

Purpose:

Logs all incoming requests.

JavaScript Code:

```
app.use((req, res, next) => {  
  console.log(req.method, req.url);  
  next();  
});
```

Validation Middleware

Validation checks:

Student number

Email format

Password strength

Error Handling Middleware

Handles application errors gracefully.

JavaScript Code:

```
app.use((err, req, res, next) => {  
  res.status(500).json({ error: err.message });  
});
```

AI INTEGRATION AND SMART FEATURES

AI Chatbot

Technology Used:
OpenRouter API with Meta Llama 3.2

Features:

Answers student questions
Programme guidance
Admission assistance
Smart suggestions

Smart Search

Features:

Real-time searching
Relevance scoring
Deep linking
Instant filtering

CRUD OPERATIONS

CRUD Matrix

Operation	Method	Endpoint
CREATE	POST	/api/register
READ	GET	/api/students
UPDATE	PUT	/api/student/:id
DELETE	DELETE	/api/booking/:id

INSTALLATION GUIDE

Step 1 — Clone Repository

Step 2 — Install Dependencies

Step 3 — Configure Database

Create database

Import

Step 4 — Start Server

TEAM CONTRIBUTIONS

Member	Role	Contribution
A.N. Mbhense	Lead Developer	Backend, AI, CRUD
A.L. Ntuli	Frontend Designer	UI/UX, responsiveness
S.N.T. Masuku	QA & Documentation	Testing and documentation

CONCLUSION

The project successfully transformed the MUT ICT Department website into a modern smart web application. The system now supports user authentication, CRUD operations, AI-powered assistance, database integration, responsive design, and interactive frontend features.

The project also demonstrated practical implementation of full-stack development technologies including Node.js, Express.js, MySQL, JavaScript, and AI APIs.

FUTURE WORK

Feature	Description
Email Verification	Verify student accounts
Password Reset	Recover forgotten passwords
Mobile App	React Native application
Analytics Dashboard	Lecturer statistics
Payment Gateway	Online fee payments

REFERENCES

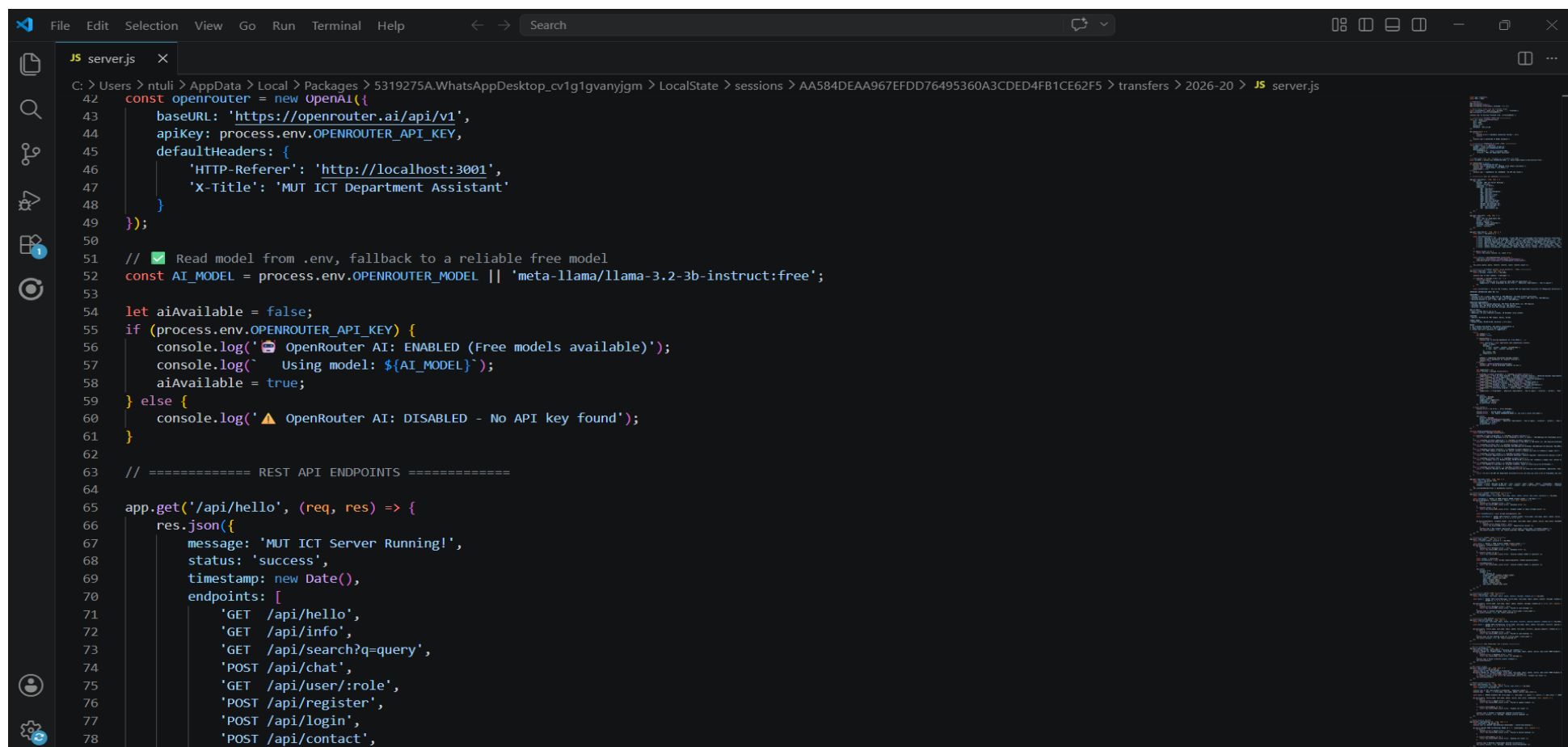
[Node.js Official Website](#)
[Express.js Documentation](#)
[MySQL Documentation](#)
[OpenRouter API](#)
[GitHub Repository](#)
[ChatGPT by OpenAI](#)

APPENDICES

Appendix A — GitHub Repository

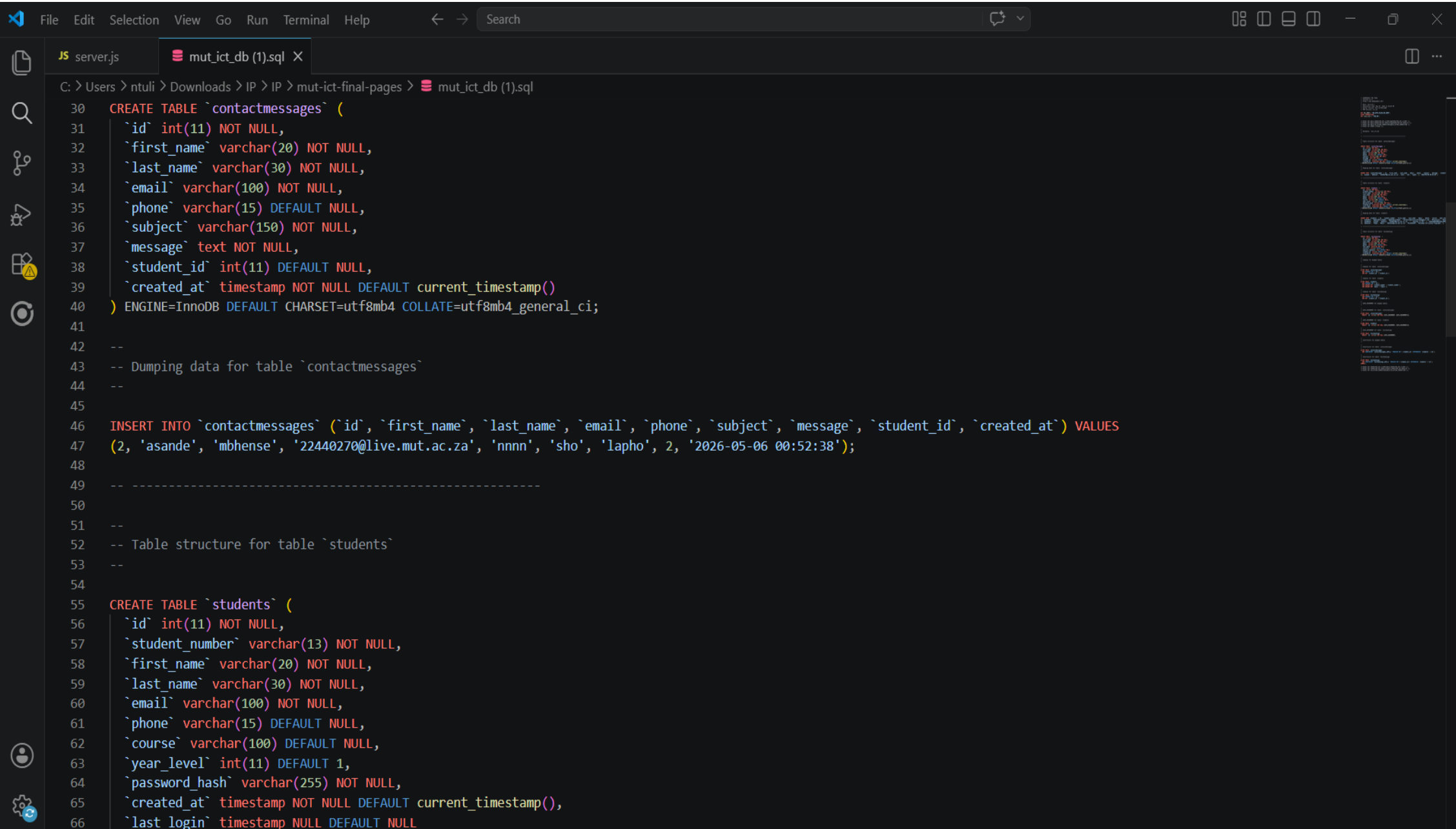
[GitHub Repository](#)

Appendix B — Server Code



```
JS server.js
C: > Users > ntuli > AppData > Local > Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > AA584DEAA967EFDD76495360A3CDED4FB1CE62F5 > transfers > 2026-20 > JS server.js
42 const openrouter = new OpenAI({
43   baseUrl: 'https://openrouter.ai/api/v1',
44   apiKey: process.env.OPENROUTER_API_KEY,
45   defaultHeaders: {
46     'HTTP-Referer': 'http://localhost:3001',
47     'X-Title': 'MUT ICT Department Assistant'
48   }
49 });
50
51 // Read model from .env, fallback to a reliable free model
52 const AI_MODEL = process.env.OPENROUTER_MODEL || 'meta-llama/llama-3.2-3b-instruct:free';
53
54 let aiAvailable = false;
55 if (process.env.OPENROUTER_API_KEY) {
56   console.log('📡 OpenRouter AI: ENABLED (Free models available)');
57   console.log(` Using model: ${AI_MODEL}`);
58   aiAvailable = true;
59 } else {
60   console.log('⚠️ OpenRouter AI: DISABLED - No API key found');
61 }
62
63 // ===== REST API ENDPOINTS =====
64
65 app.get('/api/hello', (req, res) => {
66   res.json({
67     message: 'MUT ICT Server Running!',
68     status: 'success',
69     timestamp: new Date(),
70     endpoints: [
71       'GET /api/hello',
72       'GET /api/info',
73       'GET /api/search?q=query',
74       'POST /api/chat',
75       'GET /api/user:role',
76       'POST /api/register',
77       'POST /api/login',
78       'POST /api/contact',
```

Appendix B — DataBase Code



The screenshot shows a code editor with two tabs: 'server.js' and 'mut_ict_db (1).sql'. The active tab is 'mut_ict_db (1).sql', which contains SQL code for creating two tables: 'contactmessages' and 'students'. The code is as follows:

```
30 CREATE TABLE `contactmessages` (  
31   `id` int(11) NOT NULL,  
32   `first_name` varchar(20) NOT NULL,  
33   `last_name` varchar(30) NOT NULL,  
34   `email` varchar(100) NOT NULL,  
35   `phone` varchar(15) DEFAULT NULL,  
36   `subject` varchar(150) NOT NULL,  
37   `message` text NOT NULL,  
38   `student_id` int(11) DEFAULT NULL,  
39   `created_at` timestamp NOT NULL DEFAULT current_timestamp()  
40 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_general_ci;  
41  
42 --  
43 -- Dumping data for table `contactmessages`  
44 --  
45  
46 INSERT INTO `contactmessages` (`id`, `first_name`, `last_name`, `email`, `phone`, `subject`, `message`, `student_id`, `created_at`) VALUES  
47 (2, 'asande', 'mbhense', '22440270@live.mut.ac.za', 'nnnn', 'sho', 'lapho', 2, '2026-05-06 00:52:38');  
48  
49 -----  
50  
51 --  
52 -- Table structure for table `students`  
53 --  
54  
55 CREATE TABLE `students` (  
56   `id` int(11) NOT NULL,  
57   `student_number` varchar(13) NOT NULL,  
58   `first_name` varchar(20) NOT NULL,  
59   `last_name` varchar(30) NOT NULL,  
60   `email` varchar(100) NOT NULL,  
61   `phone` varchar(15) DEFAULT NULL,  
62   `course` varchar(100) DEFAULT NULL,  
63   `year_level` int(11) DEFAULT 1,  
64   `password_hash` varchar(255) NOT NULL,  
65   `created_at` timestamp NOT NULL DEFAULT current_timestamp(),  
66   `last_login` timestamp NULL DEFAULT NULL
```